

Protect and track

Reserve for risk, phase spending over time, and check that the project still holds up against the plan.

9. Reserves

Expected monetary value (EMV) = probability × incremental cost impact. Our contingency reserve is the sum of the five risk EMVs: \$22,500 . All amounts are in USD; probabilities and recovery effort are planning assumptions, not measured failure rates.

Risk register and contingency calculation

Risk	Chance	Cost impact and basis	EMV
A key developer leaves mid-project	15%	\$80,000: additional onboarding, knowledge transfer, and rework.	\$12,000
Launch cannot handle expected workload	20%	\$30,000: unplanned capacity adjustments and recovery beyond the capacity already budgeted.	\$6,000
Premium features are not developed in time	5%	\$30,000: 2 additional FTE-months × \$15,000 for completion, integration, and regression testing. Lost revenue is not included.	\$1,500
Live feedback exceeds the latency target	15%	\$15,000: 1 additional FTE-month for profiling, buffering/rendering improvements, and latency retesting under normal load.	\$2,250
Webcam or device incompatibility	10%	\$7,500: 0.5 additional FTE-month for camera/browser compatibility fixes and device testing.	\$750
Contingency reserve: sum of EMVs			\$22,500

Risk responses and triggers

Proposed response responsibilities

Risk / proposed owner	Prevention or mitigation	Trigger for review
Developer departure / technical lead	Document critical components, share code ownership, and keep onboarding notes current.	A departure notice or an unfilled critical development responsibility.
Launch workload / infrastructure lead	Plan self-hosted capacity at three times expected launch demand for the first launch month; load-test before release. Power down extra servers if sustained demand is low.	Load tests or launch monitoring show demand exceeding tested capacity. Confirm that planned headroom fits the infrastructure allocation before launch.
Premium delay / product lead	Prioritize the core premium scope and review feature completion each sprint.	A premium milestone slips or a required feature fails acceptance testing.

Risk / proposed owner	Prevention or mitigation	Trigger for review
Latency / frontend and QA leads	Measure frame-capture-to-visible-feedback delay using simulated feedback; agree on a latency threshold before testing.	Measured delay exceeds the agreed target under normal operating load.
Compatibility / QA lead	Test supported browsers, camera permissions, and representative mobile and desktop devices early.	A supported device cannot reliably start the camera or display feedback.

Planned mitigation belongs in the base estimate; risk impacts cover additional recovery only. Capacity, latency, and compatibility costs must not be charged twice for the same work. Powering down self-hosted servers reduces avoidable running costs, not hardware costs already incurred.

Reserve policy

Contingency versus management reserve

Policy	Contingency reserve	Management reserve
Covers	Identified risks in the risk register.	Unforeseen work within the approved project scope.
Inside the cost baseline?	Yes	No
Controlled by	Project manager, within approved authority.	Project sponsor, through change control.
Sizing formula	Sum of risk EMVs = \$22,500.	5% × \$563,954 cost baseline = \$28,197.70.
Release and tracking	Record the risk, approved response amount, date, and remaining reserve.	Require sponsor approval; document the baseline change and remaining reserve.

10. Time-phase and check

Monthly labor = scheduled FTE × \$15,000 per FTE-month. Recurring non-labor is \$1,400 infrastructure + \$330 tools per month. For this schedule, the \$2,500 legal fee is placed in M1 and \$124 app integration fees in M6.

Six-month base build spending plan (USD)

Month	FTE	Labor	Non-labor	Monthly total	Cumulative total
M1	5.50	\$82,500	\$4,230	\$86,730	\$86,730
M2	5.90	\$88,500	\$1,730	\$90,230	\$176,960
M3	6.05	\$90,750	\$1,730	\$92,480	\$269,440
M4	5.95	\$89,250	\$1,730	\$90,980	\$360,420
M5	6.05	\$90,750	\$1,730	\$92,480	\$452,900
M6	5.78	\$86,700	\$1,854	\$88,554	\$541,454
Total	35.23	\$528,450	\$13,004	\$541,454	\$541,454

The monthly plan uses the rounded staffing allocations from Size the people . It phases base build costs only: reserves remain separately controlled until approved for use. Annual operating costs are separate and are not included in this build request.

Full build request

From base estimate to total funding request (USD)

Budget line	Calculation / basis	Amount
Labor	35.23 FTE-months × \$15,000	\$528,450.00
Non-labor	\$8,400 infrastructure + \$1,980 tools + \$0 incremental AI licenses + \$124 app fees + \$2,500 legal	\$13,004.00
Base build estimate	Labor + non-labor	\$541,454.00
Contingency reserve	Sum of five risk EMVs	\$22,500.00
Cost baseline	Base build estimate + contingency reserve	\$563,954.00
Management reserve	5% × cost baseline	\$28,197.70
Total build request	Cost baseline + management reserve	\$592,151.70

Budget checks and tracking

Checks before and during delivery

Check	Result or tracking rule
Staffing and spending reconcile	Monthly staffing totals 35.23 FTE-months. Monthly labor totals \$528,450 and non-labor totals \$13,004; together they match the \$541,454 base estimate.
Cost mix	Labor is 97.60% of the base estimate. Review the small non-labor allowance against the self-hosting and zero incremental AI-license assumptions.
Reserve traceability	The \$22,500 contingency equals the five EMVs. Keep a reserve log with opening balance, approved draws, linked risks, and remaining balance.
Monthly spending and delivery	At each month-end, record actual spending, compare actual minus planned spending, and review accepted work and milestones. Positive spending variance means spending above plan; lower spending alone does not prove the project is ahead.
Forecast and approvals	Update the forecast as actual cost to date + estimated remaining cost. Review overruns and reserve requests with the project manager; escalate management-reserve use to the sponsor.

This is a planning baseline, not a report of actual spending. Revisit risk probabilities, remaining work, and the launch capacity assumption at each monthly review.

Budget charts

Monthly build spending (USD)



Cumulative planned build spending (USD)



Base build costs only. Contingency and management reserves are excluded.

Monthly spending and cumulative base build costs. Both charts start at zero and exclude reserves. Select the charts to view them at full size; the tables above provide the exact values.